

How Google Cloud is Integrating AI *and* ML into its Services

This article provides an overview of the native AIML services offered by Google Cloud.



Development, training, tuning, and deployment of machine learning (ML) models are all time-consuming tasks. These can be simplified with an integrated ML based deployment pipeline activity termed as MLOps. A Google Anthos based template can help to create Juniper notebooks for building and deploying ML models using a predefined data model and a four-stage approach, as follows.

Prepare: To enable data modelling, load/ingest data for model preparation, query data sets, and create model storage in a cloud storage service.

Build: Create deployment notebooks using predefined Juniper notebooks, which contain PySpark,

conda framework or TensorFlow to quickly create a development workspace for building ML models. There are predefined containers or VM images available, which can be leveraged for quickly building an ML environment.

Validate: Use the What-if Tool (WIT) for testing the performance of ML models and Vizier black-box optimisation framework to optimise complex ML models.

Deploy: Create predictive ML models from the build templates and configuration to enable real-time deployment of ML models for production-ready activities and end user usage.

Figure 1 highlights the different stages of Google's AI platform.

Using traditional CI/CD pipelines needs a lot of scripting and coding to create templates for ML model development, preparation, build and deployment. MLOps helps to reduce this effort with a unified architecture for ML models using a no-code platform, robust governance and managed services for faster 'time to market' by leveraging the full power of Google's AI platform.

Google DocAI

For any enterprise, document processing is a complex and large volume task. Enterprise IT solutions that combine artificial intelligence (AI), ML and content management systems